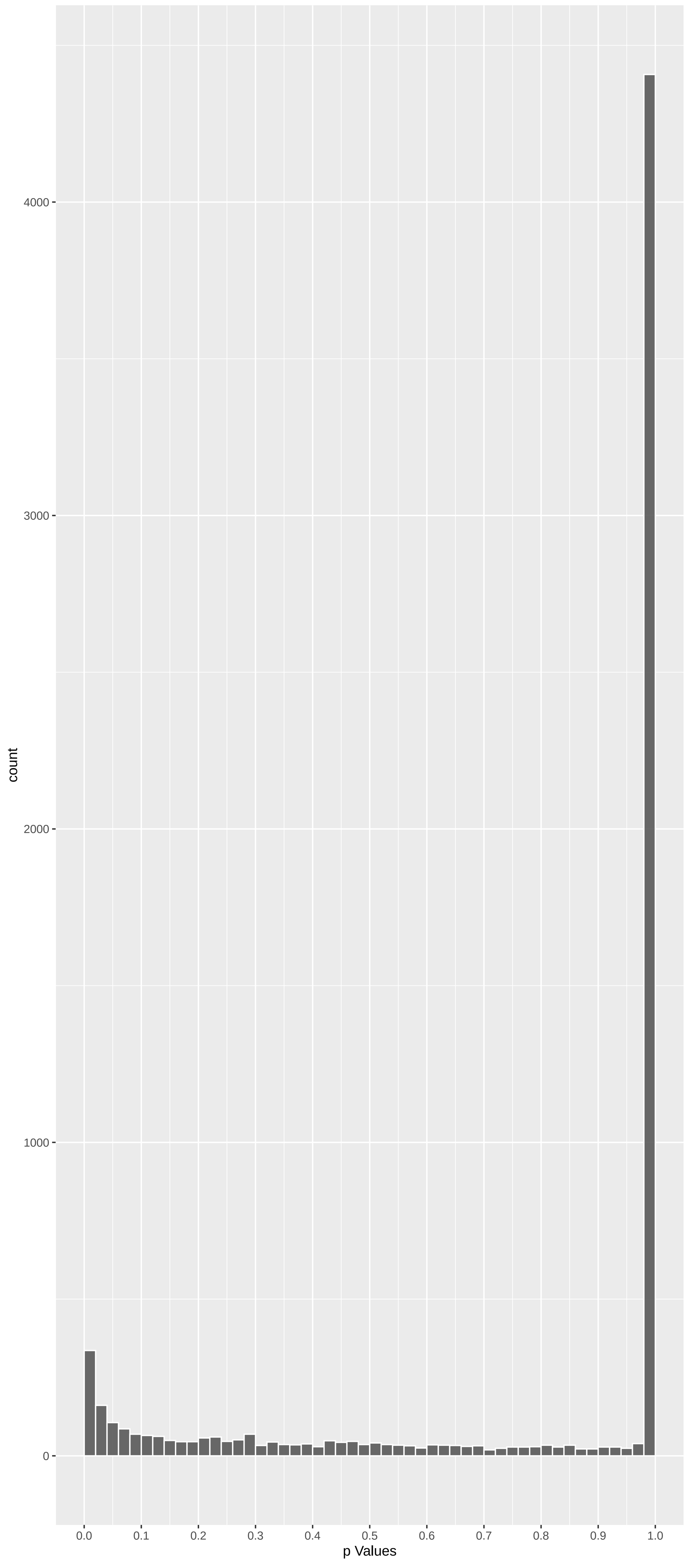
Top genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
MUC20	−5.734162	5.879751e−08	4.687e−04	1.000e+00
EP400	5.695907	7.363077e−08	4.687e−04	1.000e+00
TRAF2	5.366982	4.803897e−07	1.529e−03	1.000e+00
TET1	−5.412206	3.735173e−07	1.529e−03	1.000e+00
SLBP	5.169962	1.404849e−06	3.577e−03	1.000e+00
TGFBRAP1	4.990864	3.606586e−06	5.984e−03	1.000e+00
ZDHHC8	4.982767	3.760880e−06	5.984e−03	1.000e+00
NEK5	−5.025479	3.013069e−06	5.984e−03	1.000e+00
ZNF281	4.953692	4.369099e−06	6.180e−03	1.000e+00
SHANK3	4.907807	5.526018e−06	7.035e−03	1.000e+00
USP11	4.817036	8.742371e−06	8.561e−03	1.000e+00
PHRF1	4.834832	7.995494e−06	8.561e−03	1.000e+00
ARRDC1	4.819168	8.649473e−06	8.561e−03	1.000e+00
NEDD4	−4.760129	1.160816e−05	1.056e−02	1.000e+00
LRRC53	−4.714732	1.452172e−05	1.087e−02	1.000e+00
GOLGA4	−4.724502	1.384078e−05	1.087e−02	1.000e+00
PKD1L3	−4.722785	1.395823e−05	1.087e−02	1.000e+00
PARD6G	4.683856	1.689166e−05	1.195e−02	1.000e+00
WDCP	−4.611734	2.395941e−05	1.525e−02	1.000e+00
NSD2	4.621673	2.283949e−05	1.525e−02	1.000e+00
MCM3AP	4.578688	2.807415e−05	1.702e−02	1.000e+00
CSTF2	4.548832	3.236667e−05	1.873e−02	1.000e+00
CPT2	−4.538433	3.400434e−05	1.882e−02	1.000e+00
SEC14L5	4.498192	4.112032e−05	1.965e−02	1.000e+00
HERC2	4.467975	4.737805e−05	1.965e−02	1.000e+00

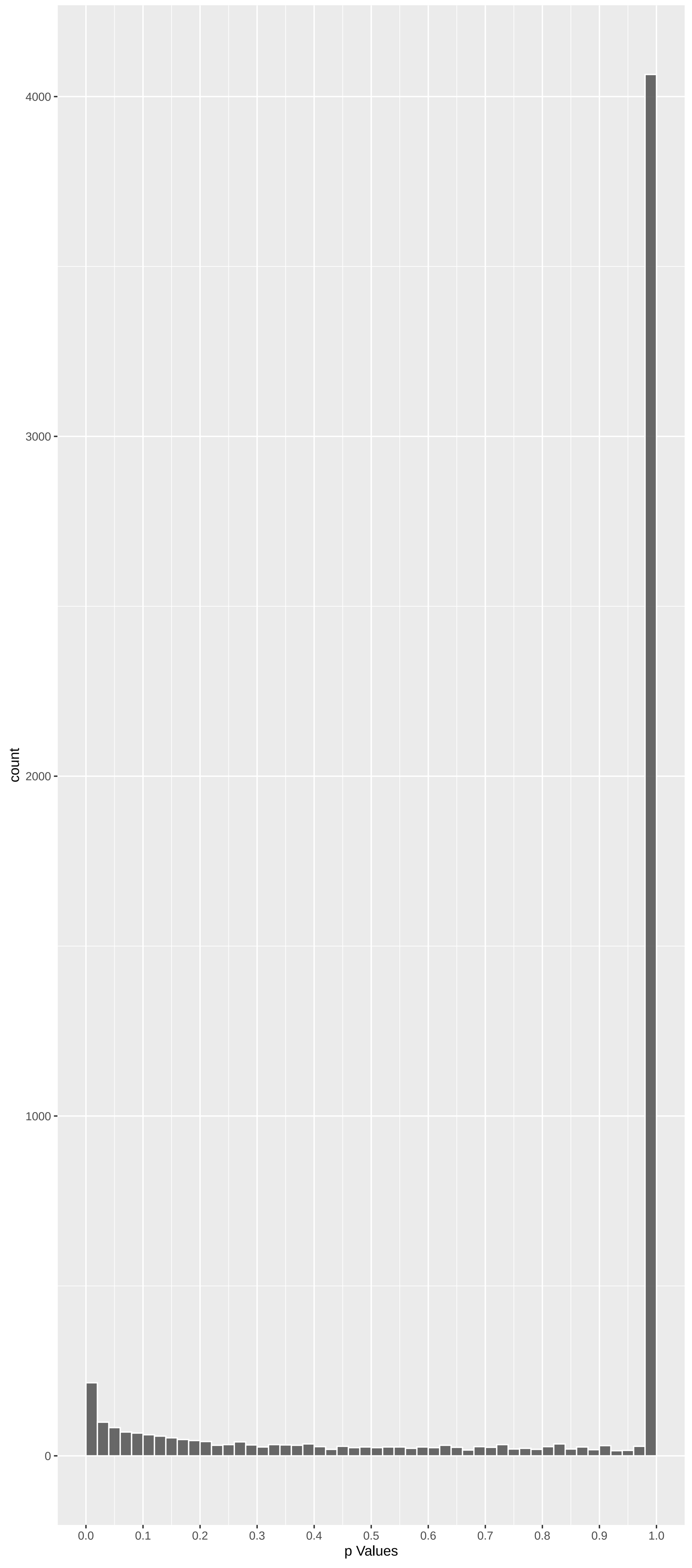
Top genes by Q−Value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
M6PR	0.942758474	1.000000000	1.000e+00	1.000e+00
FKBP4	1.637741701	0.608853526	1.000e+00	1.000e+00
NDUFAF7	−1.552652682	0.723036795	1.000e+00	1.000e+00
FUCA2	−1.103769483	1.000000000	1.000e+00	1.000e+00
DBNDD1	2.230592606	0.154248762	1.000e+00	1.000e+00
HS3ST1	1.369996856	1.000000000	1.000e+00	1.000e+00
SEMA3F	−0.738288003	1.000000000	1.000e+00	1.000e+00
CFTR	−1.245432758	1.000000000	1.000e+00	1.000e+00
CYP51A1	2.046411644	0.244295324	1.000e+00	1.000e+00
USP28	0.423611864	1.000000000	1.000e+00	1.000e+00
SLC7A2	0.053196122	1.000000000	1.000e+00	1.000e+00
PDK4	3.408292793	0.003922243	2.098e−01	1.000e+00
USH1C	−0.009842999	1.000000000	1.000e+00	1.000e+00
RALA	2.278659748	0.136123795	1.000e+00	1.000e+00
BAIAP2L1	2.678394816	0.044385578	6.632e−01	1.000e+00
CACNG3	3.107716077	0.011312346	3.470e−01	1.000e+00
TMEM132A	1.065623826	1.000000000	1.000e+00	1.000e+00
DVL2	−1.771936242	0.458430849	1.000e+00	1.000e+00
RPAP3	−0.442512616	1.000000000	1.000e+00	1.000e+00
TRAPPC6A	−0.324766341	1.000000000	1.000e+00	1.000e+00
SPATA20	−1.427567161	0.920498888	1.000e+00	1.000e+00
CD79B	−0.984253330	1.000000000	1.000e+00	1.000e+00
RHBDD2	1.102262002	1.000000000	1.000e+00	1.000e+00
RPUSD1	2.850025420	0.026229441	5.190e−01	1.000e+00
TSR3	−0.498048610	1.000000000	1.000e+00	1.000e+00

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
EP400	5.695907	7.363077e−08	4.687e−04	1.000e+00
TRAF2	5.366982	4.803897e−07	1.529e−03	1.000e+00
SLBP	5.169962	1.404849e−06	3.577e−03	1.000e+00
TGFBRAP1	4.990864	3.606586e−06	5.984e−03	1.000e+00
ZDHHC8	4.982767	3.760880e−06	5.984e−03	1.000e+00
ZNF281	4.953692	4.369099e−06	6.180e−03	1.000e+00
SHANK3	4.907807	5.526018e−06	7.035e−03	1.000e+00
USP11	4.817036	8.742371e−06	8.561e−03	1.000e+00
PHRF1	4.834832	7.995494e−06	8.561e−03	1.000e+00
ARRDC1	4.819168	8.649473e−06	8.561e−03	1.000e+00

Top Negative genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
MUC20	−5.734162	5.879751e−08	4.687e−04	1.000e+00
TET1	−5.412206	3.735173e−07	1.529e−03	1.000e+00
NEK5	−5.025479	3.013069e−06	5.984e−03	1.000e+00
NEDD4	−4.760129	1.160816e−05	1.056e−02	1.000e+00
LRRC53	−4.714732	1.452172e−05	1.087e−02	1.000e+00
GOLGA4	−4.724502	1.384078e−05	1.087e−02	1.000e+00
PKD1L3	−4.722785	1.395823e−05	1.087e−02	1.000e+00
WDCP	−4.611734	2.395941e−05	1.525e−02	1.000e+00
CPT2	−4.538433	3.400434e−05	1.882e−02	1.000e+00
CYBRD1	−4.462809	4.853521e−05	1.965e−02	1.000e+00

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NADH Dehydrogenase Complex Assembly (GO:	−0.25231564	41	2.340e−08	6.285e−05	TMEM126A:20 TMEM126B:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223
Mitochondrial Respiratory Chain Complex	−0.25231564	41	2.340e−08	6.285e−05	TMEM126A:20 TMEM126B:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223
Mitochondrial Respiratory Chain Complex	−0.19480931	62	1.179e−07	2.110e−04	TMEM126A:20 TMEM126B:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223
Mitochondrial Gene Expression (GO:014005	−0.15155407	91	6.201e−07	8.326e−04	PTCD3:27 MRPL37:33 MRPL43:176 RARS2:225 MRPS30:226 FASTKD5:290
Mitochondrial Translation (GO:0032543)	−0.15186334	87	1.034e−06	1.111e−03	PTCD3:27 MRPL37:33 MTIF2:175 MRPL43:176 RARS2:225 MRPS30:226
tRNA Methylation (GO:0030488)	−0.21803620	34	1.101e−05	9.856e−03	TRMT9B:145 TRMT10C:184 METTL8:444 THUMPD2:483 TYW3:517 TRMT10B:598
Oxidative Phosphorylation (GO:0006119)	−0.18085227	41	6.255e−05	4.799e−02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 ATP5PB:368 TEFM:423
Mitochondrial Electron Transport, NADH T	−0.22550405	25	9.591e−05	6.439e−02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 NDUFB4:424 NDUFA8:464
Aerobic Respiration (GO:0009060)	−0.17835275	39	1.178e−04	7.033e−02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 TEFM:423 NDUFB4:424
Cell Junction Assembly (GO:0034329)	0.13896227	63	1.398e−04	7.510e−02	SHANK3:7 SHANK2:62 SDK1:143 ARVCF:147 DBNL:169 SDK2:368
Calcium Ion Transport (GO:0006816)	0.11348239	93	1.613e−04	7.875e−02	ITPR1:24 TRPM5:49 TRPM1:107 CACNA1S:201 CACNG3:266 CRACR2B:285
Proton Motive Force−Driven Mitochondrial	−0.17965487	36	1.937e−04	8.670e−02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 ATP5PB:368 NDUFB4:424
tRNA Modification (GO:0006400)	−0.13657067	61	2.301e−04	9.508e−02	ALKBH8:116 TRMT9B:145 METTL8:444 THUMPD2:483 DTWD1:504 TYW3:517
Proton Motive Force−Driven ATP Synthesis	−0.17174606	38	2.520e−04	9.666e−02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 ATP5PB:368 NDUFB4:424
Cilium Assembly (GO:0060271)	−0.07339778	197	4.111e−04	1.472e−01	ARL13B:21 CEP126:24 FAM161A:31 TCTN3:40 GORAB:66 DNMBP:78
Calcium Ion Import Across Plasma Membran	0.17509270	33	5.046e−04	1.694e−01	CACNA1B:47 SCN2A:81 TRPM1:107 CACNA1S:201 TRPV6:300 CLU:396
mRNA 3'−End Processing (GO:0031124)	0.18425382	28	7.449e−04	2.278e−01	SLBP:3 CSTF2:14 CDC73:73 WDR33:304 SYMPK:386 PEX2:508
Muscle Contraction (GO:0006936)	0.11492762	72	7.635e−04	2.278e−01	CACNA1S:201 LMOD1:245 MYH2:322 EMD:399 MYH13:819 RYR2:833
Aerobic Electron Transport Chain (GO:001	−0.14145558	43	1.345e−03	2.535e−01	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 NDUFB4:424 NDUFA8:464
Cell−Cell Junction Assembly (GO:0007043)	0.12854013	53	1.226e−03	2.535e−01	ARVCF:147 JAM3:275 TJP1:478 TLN1:514 GJA4:584 TBCD:751
Cellular Respiration (GO:0045333)	−0.12947663	53	1.128e−03	2.535e−01	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 NDUFB4:424 NDUFA8:464
Cilium Organization (GO:0044782)	−0.06895408	183	1.369e−03	2.535e−01	ARL13B:21 CEP126:24 FAM161A:31 TCTN3:40 DNMBP:78 CCDC66:105
Glycogen Metabolic Process (GO:0005977)	0.23932592	15	1.335e−03	2.535e−01	PYGB:101 PER2:268 PYGM:597 GYS1:1044 GYG2:1104 GYS2:1280
Mitochondrial ATP Synthesis Coupled Elec	−0.14145558	43	1.345e−03	2.535e−01	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 NDUFB4:424 NDUFA8:464
Mitochondrial RNA Processing (GO:0000963	−0.31631748	9	1.018e−03	2.535e−01	SUPV3L1:46 TRMT10C:184 FASTKD5:290 PNPT1:511 FASTKD2:1042 FASTKD1:1105
Negative Regulation Of Fibroblast Prolif	0.28705643	11	9.806e−04	2.535e−01	CDC73:73 MYC:338 EMD:399 PEX2:508 TRIM32:1520 SKI:1905
Negative Regulation Of Transcription By	0.04598413	440	1.090e−03	2.535e−01	ZNF281:6 EHMT1:33 ZBTB45:39 TCF25:54 TCFL5:59 CDC73:73
Positive Regulation Of Cytosolic Calcium	0.20811836	20	1.279e−03	2.535e−01	OPRL1:112 F2RL1:217 GPR55:308 GRM1:418 GPR35:465 GPR20:510
tRNA Wobble Base Modification (GO:000209	−0.27510614	12	9.699e−04	2.535e−01	ALKBH8:116 TRMT9B:145 ELP2:182 ELP5:815 ELP4:886 MTO1:1010
DNA Repair (GO:0006281)	−0.05999278	234	1.683e−03	3.013e−01	BRCA1:16 RNF169:49 FANCM:81 RNF168:115 DCLRE1A:125 HROB:144
Chemical Synaptic Transmission (GO:00072	0.06556424	188	2.040e−03	3.402e−01	CACNA1B:47 RIMBP2:76 GABRB3:78 KCNQ3:172 TH:254 RPS6KA2:299
mRNA Export From Nucleus (GO:0006406)	0.14680814	37	2.018e−03	3.402e−01	MCM3AP:13 IWS1:230 NUP160:372 NUP133:442 ALYREF:592 THOC5:634
Regulation Of Cilium Movement (GO:000335	−0.33490171	7	2.154e−03	3.402e−01	DNAH11:203 CFAP206:361 CFAP43:704 CYB5D1:725 CCDC65:820 CCDC40:6080
Synaptic Vesicle Uncoating (GO:0016191)	0.44346531	4	2.129e−03	3.402e−01	GAK:86 SH3GL2:498 SYNJ1:1067 SH3GL3:1101 NA NA
Regulation Of Long−Term Synaptic Potenti	0.20185110	19	2.328e−03	3.573e−01	SHANK3:7 ADCY1:25 NSG1:314 SHISA7:1002 YTHDF1:1346 ADCY8:1347
poly(A)+ mRNA Export From Nucleus (GO:00	0.27720566	10	2.406e−03	3.589e−01	MCM3AP:13 IWS1:230 NUP133:442 ZC3H11A:1088 GLE1:1805 DDX25:2097
RNA Export From Nucleus (GO:0006405)	0.14017085	38	2.814e−03	3.637e−01	MCM3AP:13 NUP160:372 NUP133:442 ALYREF:592 NUP155:633 THOC5:634
Adenylate Cyclase−Inhibiting G Protein−C	0.14121403	37	2.979e−03	3.637e−01	OPRL1:112 GNAI3:239 HRH3:356 OPRK1:625 PDE2A:717 DRD2:1061
Adherens Junction Organization (GO:00343	0.16239026	28	2.956e−03	3.637e−01	JAM3:275 INAVA:369 TJP1:478 MTSS1:583 TBCD:751 CDH6:781
Cell−Cell Junction Organization (GO:0045	0.12348637	49	2.819e−03	3.637e−01	TGFB1:89 ARVCF:147 F2RL1:217 TJP1:478 TLN1:514 GJA4:584

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	0.30081687	32	3.934e−09	2.548e−05	INTS1:71 ADAP1:154 EIF3B:181 TTYH3:249 GRIFIN:301 MAFK:353
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	−0.25124301	38	8.478e−08	2.745e−04	TMEM126B:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223 NDUFB4:424
BENPORATH_ES_WITH_H3K27ME3	0.05798049	733	1.329e−07	2.870e−04	RGS9:26 FRMPD4:42 CACNA1B:47 MMP2:48 SMPD3:58 COL12A1:75
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	0.39551650	14	3.001e−07	4.169e−04	EP400:1 P2RX2:178 POLE:214 GOLGA3:378 ANKLE2:425 ULK1:578
REACTOME_COMPLEX_I_BIOGENESIS	−0.23659871	39	3.219e−07	4.169e−04	TMEM126B:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223 NDUFA9:353
REACTOME_DEVELOPMENTAL_BIOLOGY	0.05709612	679	5.420e−07	5.850e−04	MMP2:48 ARHGAP39:52 PTK2:63 TRIO:69 ABLIM2:72 SCN2A:81
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	−0.15805506	79	1.230e−06	1.138e−03	LRPPRC:42 TMEM126B:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223
PEREZ_TP53_TARGETS	0.05013360	813	1.694e−06	1.371e−03	TGFBRAP1:4 PHRF1:8 SEC14L5:15 MICAL3:18 BRD1:19 ADCY1:25
NIKOLSKY_BREAST_CANCER_5P15_AMPLICON	0.30743973	20	1.947e−06	1.401e−03	EXOC3:61 CEP72:191 LRRC14B:566 IRX4:621 BRD9:741 SLC6A18:884
REACTOME_INFECTIOUS_DISEASE	0.05854877	555	3.013e−06	1.774e−03	ZDHHC8:5 ENTPD1:22 ITPR1:24 ADCY1:25 PTK2:63 NUP37:68
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	−0.16680093	66	2.848e−06	1.774e−03	LRPPRC:42 TMEM126B:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223
JOHNSTONE_PARVB_TARGETS_3_DN	−0.05761698	568	3.374e−06	1.821e−03	RTKN2:59 PFKFB3:111 ARHGAP11A:114 ZWINT:118 SPAG5:122 LRR1:142
REACTOME_NEURONAL_SYSTEM	0.08110099	270	4.978e−06	2.480e−03	ADCY1:25 CACNA1B:47 SHANK2:62 GABRB3:78 DBNL:169 KCNQ3:172
GINESTIER_BREAST_CANCER_ZNF217_AMPLIFIED	0.07995508	253	1.299e−05	6.009e−03	TRAF2:2 BRD1:19 INTS1:71 SPON2:84 EPN1:140 SH3BP2:145
WP_2Q37_COPY_NUMBER_VARIATION_SYNDROME	0.13416109	86	1.749e−05	7.550e−03	KIF1A:31 KLHL30:57 TRAF3IP1:106 DGKD:115 GPC1:116 MROH2A:126
MOOTHA_HUMAN_MITODB_6_2002	−0.06851299	328	2.225e−05	9.005e−03	CPT2:9 SUPV3L1:46 ACADM:57 NDUFS3:77 NDUFB8:95 MLH3:147
REACTOME_MITOCHONDRIAL_TRANSLATION	−0.13495764	82	2.451e−05	9.336e−03	PTCD3:27 MRPL37:33 MTIF2:175 MRPL43:176 MRPS30:226 ERAL1:378
GEORGES_TARGETS_OF_MIR192_AND_MIR215	−0.04845097	655	2.908e−05	1.046e−02	WDCP:8 BRCA1:16 ATAD5:36 NIN:38 DST:54 RTKN2:59
WP_OXIDATIVE_PHOSPHORYLATION	−0.20296600	35	3.274e−05	1.116e−02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 ATP5PB:368 NDUFB4:424
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.05814282	435	3.684e−05	1.142e−02	CADM3:32 ADGRB1:53 OTUD7A:104 MAPK8IP2:111 KCNQ3:172 CNTN4:175
REACTOME_TRANSPORT_OF_MATURE_TRANSCRIPT_	0.15809903	57	3.705e−05	1.142e−02	SLBP:3 NUP37:68 NUP210:164 WDR33:304 NUP43:351 NUP160:372
NIKOLSKY_BREAST_CANCER_16P13_AMPLICON	0.13206946	78	5.633e−05	1.658e−02	CRAMP1:80 UNKL:132 JPT2:267 CAPN15:328 RPUSD1:391 MEIOB:419
MARTENS_TRETINOIN_RESPONSE_UP	0.05008692	558	6.194e−05	1.744e−02	KIF1A:31 GJB4:34 SHANK2:62 SPON2:84 WNK2:97 CHST15:125
MIKKELSEN_NPC_ICP_WITH_H3K4ME3	−0.06590007	311	7.039e−05	1.850e−02	WDCP:8 KLHL5:23 A2M:25 ALKBH8:116 DCLRE1A:125 RBMS2:135
REACTOME_PROCESSING_OF_CAPPED_INTRON_CON	0.08880110	169	7.141e−05	1.850e−02	SLBP:3 CSTF2:14 NUP37:68 NUP210:164 SUGP1:282 WDR33:304
REACTOME_DISEASES_OF_DNA_REPAIR	−0.16662397	47	7.833e−05	1.951e−02	BRCA1:16 RAD51AP1:170 ATM:244 MUTYH:362 RMI1:429 WRN:438
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.07455828	230	1.043e−04	2.501e−02	CADM3:32 ADGRB1:53 STUM:60 PTGFRN:88 TGFB1:89 GJA3:144
MIKKELSEN_NPC_HCP_WITH_H3K27ME3	0.07268223	241	1.087e−04	2.513e−02	CADM3:32 ADGRB1:53 SLCO4C1:182 CPZ:187 TFAP2D:236 KCNQ2:343
REACTOME_NERVOUS_SYSTEM_DEVELOPMENT	0.05711356	389	1.228e−04	2.743e−02	MMP2:48 ARHGAP39:52 PTK2:63 TRIO:69 ABLIM2:72 SCN2A:81
REACTOME_SIGNALING_BY_RECEPTOR_TYROSINE_	0.05789260	376	1.285e−04	2.774e−02	ITPR1:24 PTK2:63 GABRB3:78 EPN1:140 PTPN6:141 ADAP1:154
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	0.11045687	100	1.385e−04	2.894e−02	TCFL5:59 OPRL1:112 CABLES2:137 ZBTB46:186 KCNG1:212 SYCP2:229
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.17722494	37	1.925e−04	3.896e−02	SLBP:3 CSTF2:14 WDR33:304 SYMPK:386 SLU7:472 ALYREF:592
REACTOME_TRANSPORT_OF_MATURE_MRNAS_DERIV	0.19191232	31	2.184e−04	4.286e−02	SLBP:3 NUP37:68 NUP210:164 WDR33:304 NUP43:351 NUP160:372
BENPORATH_EED_TARGETS	0.04194902	676	2.389e−04	4.421e−02	PARP12:23 FRMPD4:42 CACNA1B:47 COL12A1:75 DGKI:95 ZCCHC2:108
EPPERT_PROGENITOR	−0.10402122	105	2.364e−04	4.421e−02	LRPPRC:42 SORD:97 ZWINT:118 HAUS6:289 DLC1:350 NET1:391
REACTOME_DNA_DOUBLE_STRAND_BREAK_REPAIR	−0.09957657	113	2.623e−04	4.719e−02	BRCA1:16 RNF168:115 RAD51AP1:170 POLQ:213 MUS81:243 ATM:244
MOOTHA_MITOCHONDRIA	−0.05747165	337	3.125e−04	5.470e−02	SUPV3L1:46 ACADM:57 METTL17:61 NDUFS3:77 NDUFB8:95 NDUFB5:158
WP_DNA_REPAIR_PATHWAYS_FULL_NETWORK	−0.10479778	98	3.443e−04	5.717e−02	BRCA1:16 FANCM:81 FANCE:211 ATM:244 MUTYH:362 XRCC4:364
WONG_MITOCHONDRIA_GENE_MODULE	−0.08469153	151	3.397e−04	5.717e−02	LRPPRC:42 NDUFS3:77 NDUFB8:95 NDUFB5:158 MRPS30:226 NDUFA9:353
KEGG_CALCIUM_SIGNALING_PATHWAY	0.08222773	152	4.828e−04	7.817e−02	ITPR1:24 ADCY1:25 CACNA1B:47 P2RX2:178 ADCY9:179 PTK2B:193

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Mitochondrial Diseases	−0.07709801	308	3.918e−06	3.827e−02	CPT2:9 SARS2:19 TMEM126A:20 LRPPRC:42 TMEM126B:55 METTL17:61
Increased CSF lactate	−0.17700443	44	4.944e−05	2.414e−01	LRPPRC:42 TMEM126B:55 NDUFS3:77 TRMT10C:184 RARS2:225 NDUFA9:353
Autism Spectrum Disorders	0.05744652	369	1.749e−04	4.271e−01	SHANK3:7 HERC2:21 ITPR1:24 EHMT1:33 SHANK2:62 JAKMIP1:67
Schizophrenia	0.03284286	1277	1.431e−04	4.271e−01	ZDHHC8:5 SHANK3:7 BRD1:19 HERC2:21 ADCY1:25 RGS9:26
Cataract, Pulverulent	0.43837711	5	6.871e−04	5.965e−01	GJA3:144 BFSP2:205 GJA8:635 CRYBB1:900 HSF4:1858 NA
Acute necrotizing encephalopathy	−0.25728441	16	3.679e−04	5.965e−01	TMEM126B:55 NDUFS3:77 FOXRED1:480 TIMMDC1:591 NDUFA11:642 NDUFAF4:695
Genu recurvatum	0.27460241	13	6.093e−04	5.965e−01	CTDP1:74 COL1A2:462 CANT1:820 FBN1:882 COL5A1:1241 XYLT1:1391
Horizontal eyebrow	0.36758899	7	7.580e−04	5.965e−01	DEAF1:849 RERE:956 EBF3:1474 PRDM16:1571 SKI:1905 TRAPPC9:2338
Hyperechogenic kidneys	−0.43451588	5	7.661e−04	5.965e−01	SARS2:19 CEP290:328 TCTN2:503 INVS:1540 REN:1864 NA
Malignant neoplasm of fallopian tube	−0.35300461	8	5.458e−04	5.965e−01	BRCA1:16 CASC1:241 ATM:244 NBR1:499 BARD1:779 BRCA2:1199
NADH:Q(1) Oxidoreductase deficiency	−0.21186539	22	5.849e−04	5.965e−01	TMEM126B:55 NDUFS3:77 FOXRED1:480 TIMMDC1:591 NDUFA11:642 NDUFAF4:695
Pallor of optic disc	−0.14312575	46	7.938e−04	5.965e−01	TMEM126A:20 ARL13B:21 FAM161A:31 TMEM126B:55 NDUFS3:77 CERKL:323
Progressive macrocephaly	−0.23926967	19	3.070e−04	5.965e−01	TMEM126B:55 NDUFS3:77 FOXRED1:480 TIMMDC1:591 NDUFA11:642 NDUFAF4:695
Aneurysm, Ruptured	0.36124569	7	9.344e−04	6.519e−01	RNH1:231 PPP1R13L:644 NOS3:646 SHC3:959 FN1:1239 ACE:1710
Central neuroblastoma	0.02983452	1085	1.290e−03	7.414e−01	HERC2:21 ITPR1:24 KIF1A:31 ZNF236:41 MMP2:48 PTK2:63
Huntington Disease	0.04723750	403	1.285e−03	7.414e−01	FAM193A:20 ITPR1:24 APC2:36 MMP2:48 TGFB1:89 HSF1:138
Slender finger	0.29552897	10	1.214e−03	7.414e−01	COL12A1:75 COL6A1:165 COL6A2:255 ZNF592:969 LFNG:1177 KANSL1:1448
Non−obstructive azoospermia	0.12889483	51	1.470e−03	7.977e−01	ENTPD6:29 BOLL:206 WDR11:295 DAZL:359 KITLG:395 MEIOB:419
CATARACT, COPPOCK−LIKE	0.36324971	5	4.910e−03	8.476e−01	GJA3:144 BFSP2:205 GJA8:635 CRYBB1:900 ISG20:6651 NA
COLORBLINDNESS, PARTIAL, DEUTAN SERIES	0.17540501	24	2.948e−03	8.476e−01	TGFB1:89 ANXA6:237 LMOD1:245 PLXNA4:630 CCR5:667 OPN1MW:1091
Epilepsy, Temporal Lobe	0.07238175	122	5.910e−03	8.476e−01	GABRB3:78 SCN2A:81 LETM1:150 KCNQ3:172 BIN1:207 SOD1:302
Hyperkinesia, Generalized	0.18619478	21	3.151e−03	8.476e−01	ADCY1:25 TH:254 CNTNAP2:528 NGF:564 CRHR1:1010 DRD2:1061
Psychoses, Drug	0.22233340	15	2.877e−03	8.476e−01	RGS9:26 ESR1:317 HTR1A:1265 SLC6A3:1411 SOD2:1584 ACE:1710
Psychoses, Substance−Induced	0.25400445	12	2.318e−03	8.476e−01	RGS9:26 ESR1:317 HTR1A:1265 SLC6A3:1411 SOD2:1584 ACE:1710
Abnormal heart beat	−0.10316780	60	5.780e−03	8.476e−01	CPT2:9 PHYH:71 PEX6:88 SLC19A2:236 SYNE2:296 LARS2:390
Abnormal mitochondria in muscle tissue	−0.19192911	22	1.840e−03	8.476e−01	TMEM126B:55 NDUFS3:77 FOXRED1:480 TIMMDC1:591 NDUFA11:642 NDUFAF4:695
Attention deficit hyperactivity disorder	0.05216824	287	2.544e−03	8.476e−01	SHANK3:7 HERC2:21 TRIO:69 MC4R:79 ARVCF:147 SORCS2:163
Autistic Disorder	0.03805871	482	4.726e−03	8.476e−01	SHANK3:7 EHMT1:33 FRMPD4:42 SHANK2:62 JAKMIP1:67 GABRB3:78
Bleeding tendency	−0.11883687	46	5.343e−03	8.476e−01	THPO:166 MPL:258 KAT6B:513 NBEAL2:519 F7:718 A2ML1:767
Carcinogenesis	0.01863596	2559	4.702e−03	8.476e−01	EP400:1 TRAF2:2 PHRF1:8 NSD2:12 CSTF2:14 APC2:36
Cardiac conduction abnormality	−0.10191300	61	5.979e−03	8.476e−01	CPT2:9 PHYH:71 PEX6:88 SLC19A2:236 SYNE2:296 LARS2:390
Catatonia	0.35270672	6	2.773e−03	8.476e−01	MLC1:294 CNP:529 SLC30A4:786 DLL4:1118 PANX2:1631 PRODH:6651
Chromosome 2q37 deletion syndrome	0.35442964	5	6.059e−03	8.476e−01	HDAC4:364 PASK:413 GPR35:465 HDLBP:1206 FARP2:6651 NA
Comatose	−0.10653395	62	3.769e−03	8.476e−01	CPT2:9 TMEM126B:55 ACADM:57 NDUFS3:77 HMGCL:421 FOXRED1:480
Common acute lymphoblastic leukemia	0.20121378	19	2.403e−03	8.476e−01	TMED2:243 KRT20:443 GJA8:635 CD38:744 MME:762 PBX1:1166
Cystic renal dysplasia	−0.30254902	9	1.674e−03	8.476e−01	NEK8:245 CEP290:328 BMPER:412 TBX18:437 MKS1:1530 TXNRD1:1682
Dementia	0.04602204	302	6.351e−03	8.476e−01	TSHZ1:66 TGFB1:89 SORL1:189 PTK2B:193 BIN1:207 F2RL1:217
ECG abnormality	−0.10316780	60	5.780e−03	8.476e−01	CPT2:9 PHYH:71 PEX6:88 SLC19A2:236 SYNE2:296 LARS2:390
Enlarged kidney	−0.21764437	15	3.526e−03	8.476e−01	CPT2:9 PKHD1:159 NEK8:245 BMPER:412 FAH:1013 GPC3:1386
Epistaxis	−0.13112685	42	3.308e−03	8.476e−01	PTPN22:180 MPL:258 NBEAL2:519 F7:718 ITGA2B:805 TET2:902

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0003635 abnormal synaptic transmissio	0.06046609	317	3.317e−04	1.579e−01	ZDHHC8:5 ENTPD1:22 ITPR1:24 ADCY1:25 CACNA1B:47 IQSEC1:51
MP0002066 abnormal motor capabilities/c	0.03344382	842	2.568e−03	2.772e−01	HERC2:21 ITPR1:24 KIF1A:31 EHMT1:33 CACNA1B:47 NEB:55
MP0009763 increased sensitivity to	−0.06231623	203	2.682e−03	2.772e−01	BRCA1:16 MBOAT4:63 DCLRE1A:125 SLC19A2:236 MUS81:243 ATM:244
MP0003252 abnormal bile duct	−0.19022372	20	3.285e−03	2.772e−01	PKHD1:159 MAP3K14:431 AHR:510 P2RY13:728 PKD2:827 ABCG5:977
MP0002063 abnormal learning/memory/cond	0.05647591	268	1.927e−03	2.772e−01	ZDHHC8:5 ADCY1:25 EHMT1:33 CACNA1B:47 MAPK8IP2:111 OPRL1:112
MP0002064 seizures	0.06149733	197	3.494e−03	2.772e−01	ITPR1:24 GABRB3:78 DGKD:115 DBNL:169 KCNQ3:172 NETO1:185
MP0001800 abnormal humoral immune	−0.04421786	290	1.177e−02	5.603e−01	NEDD4:4 ARHGEF6:161 PTPN22:180 POLQ:213 NFKBIZ:229 ATM:244
MP0002723 abnormal immune serum	−0.03349915	545	1.135e−02	5.603e−01	NEDD4:4 SKAP1:84 ARHGEF6:161 PTPN22:180 LGALS1:207 POLQ:213
MP0005025 abnormal response to	−0.04370579	311	1.012e−02	5.603e−01	HPS5:26 JAK3:291 MPO:373 GRN:408 MAP3K14:431 WRN:438
MP0009745 abnormal behavioral response	0.06112101	161	8.399e−03	5.603e−01	EHMT1:33 CACNA1B:47 GABRB3:78 OPRL1:112 PER2:268 SOD1:302
MP0002398 abnormal bone marrow	−0.03001821	655	1.426e−02	5.930e−01	NEDD4:4 BRCA1:16 HPS5:26 SUPV3L1:46 DST:54 SKAP1:84
MP0000689 abnormal spleen morphology	−0.03477712	456	1.500e−02	5.930e−01	SLC2A10:39 PKHD1:159 PTPN22:180 TMPRSS6:230 MPL:258 GHR:286
MP0008875 abnormal xenobiotic pharmacok	−0.19282672	13	1.619e−02	5.930e−01	AHR:510 ABCC3:773 CYP1A1:1137 NCEH1:1158 CYP1A2:1548 AS3MT:1754
MP0000604 amyloidosis	0.10869734	30	3.988e−02	6.145e−01	SORL1:189 CLU:396 PSEN1:591 MME:762 SORCS1:778 ZDHHC13:915
MP0005369 muscle phenotype	0.04863102	167	3.283e−02	6.145e−01	ENTPD1:22 TGFB1:89 RPTOR:91 SH3BP2:145 COL6A1:165 CACNA1S:201
MP0002452 abnormal antigen presenting	−0.03189157	489	2.145e−02	6.145e−01	NEDD4:4 ARHGEF6:161 PTPN22:180 POLQ:213 NFKBIZ:229 ATM:244
MP0002095 abnormal skin pigmentation	−0.09822458	39	3.443e−02	6.145e−01	BRCA1:16 HPS5:26 DST:54 ADAMTS20:352 ANK1:1055 POLH:1225
MP0002090 abnormal vision	0.09797271	36	4.260e−02	6.145e−01	TRPM1:107 BFSP2:205 ELOVL4:250 MERTK:272 AXL:692 CTSD:953
MP0002572 abnormal emotion/affect behav	0.04047591	228	3.938e−02	6.145e−01	ZDHHC8:5 ADCY1:25 EHMT1:33 CACNA1B:47 GABRB3:78 ADORA1:92
MP0000703 abnormal thymus morphology	−0.03923561	274	2.949e−02	6.145e−01	BRCA1:16 SUPV3L1:46 CHRD:146 AFF1:187 ATM:244 JAK3:291
MP0003315 abnormal perineum morphology	−0.15868214	14	4.006e−02	6.145e−01	GNRHR:474 FOXP3:888 SALL4:1356 GPC3:1386 PLAU:1422 RB1:6080
MP0002928 abnormal bile duct	−0.12100350	26	3.312e−02	6.145e−01	PKHD1:159 GPBAR1:400 MAP3K14:431 CHUK:1107 ABCB4:1168 MKS1:1530
MP0002282 abnormal trachea morphology	−0.10833139	30	4.056e−02	6.145e−01	CHRD:146 ARSB:250 BMPER:412 TEKT2:726 MGP:1041 PLAU:1422
MP0003091 abnormal cell migration	−0.07881540	57	4.062e−02	6.145e−01	BRCA1:16 FZR1:267 SYNE2:296 FLNB:490 GAB1:880 ITGB1:883
MP0000716 abnormal immune system	−0.02554131	811	2.320e−02	6.145e−01	NEDD4:4 BRCA1:16 SUPV3L1:46 SKAP1:84 ARHGEF6:161 PTPN22:180
MP0003690 abnormal glial cell	−0.11710660	30	2.683e−02	6.145e−01	PHYH:71 ATM:244 GRN:408 PLP1:936 GFAP:1612 PTPN11:6080
MP0003699 abnormal female reproductive	−0.03540615	338	3.043e−02	6.145e−01	BRCA1:16 PADI6:62 MLH3:147 AFF1:187 MSH4:188 TMPRSS6:230
MP0008007 abnormal cellular replicative	−0.12494930	23	3.845e−02	6.145e−01	ITGB1:883 CDK5RAP2:967 SCMH1:1129 BRCA2:1199 CUL7:1384 SKIL:1688
MP0002722 abnormal immune system	−0.02892335	506	3.425e−02	6.145e−01	BRCA1:16 SUPV3L1:46 PKHD1:159 PTPN22:180 AFF1:187 TMPRSS6:230
MP0002429 abnormal blood cell	−0.02455405	988	1.892e−02	6.145e−01	NEDD4:4 BRCA1:16 HPS5:26 SUPV3L1:46 SKAP1:84 ARHGEF6:161
MP0002971 abnormal brown adipose	0.07303726	67	3.992e−02	6.145e−01	RPTOR:91 UCP1:198 CACNA1S:201 GRM1:418 LPIN1:495 PRLH:649
MP0010094 abnormal chromosome stability	−0.07756391	58	4.213e−02	6.145e−01	BRCA1:16 POLQ:213 MUS81:243 ATM:244 XRCC4:364 BUB1B:375
MP0002019 abnormal tumor incidence	−0.03644909	337	2.607e−02	6.145e−01	BRCA1:16 FANCM:81 ITIH4:106 CASC1:241 MUS81:243 ATM:244
MP0005384 cellular phenotype	−0.03211263	346	4.723e−02	6.424e−01	BRCA1:16 DCLRE1A:125 GUSB:194 ATM:244 FUZ:246 ARSB:250
MP0002106 abnormal muscle physiology	0.05168614	128	4.603e−02	6.424e−01	ENTPD1:22 TGFB1:89 RPTOR:91 SH3BP2:145 COL6A1:165 CACNA1S:201
MP0000598 abnormal liver morphology	−0.02815909	391	6.590e−02	7.647e−01	CYBRD1:10 PHYH:71 STX2:82 PKHD1:159 GHR:286 SERPINC1:293
MP0005387 immune system phenotype	−0.05577260	89	7.130e−02	7.647e−01	SKAP1:84 C3AR1:255 JAK3:291 MAP3K14:431 TRAF3IP2:467 AHR:510
MP0005389 reproductive system phenotype	−0.06745864	60	7.229e−02	7.647e−01	TRPM6:12 SLC19A2:236 ATM:244 PTGFR:472 CAMK4:1198 CHST8:1456
MP0008058 abnormal DNA repair	−0.09101349	34	6.711e−02	7.647e−01	XRCC4:364 TEX15:449 POLL:493 POLK:561 NHEJ1:796 BRCA2:1199
MP0005551 abnormal eye electrophysiolog	0.05194434	110	6.249e−02	7.647e−01	RGS9:26 PDE6B:99 TRPM1:107 ELOVL4:250 MERTK:272 VSX1:348

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
cerebellum	0.06100445	290	4.505e−04	1.216e−02	SHANK3:7 RNF112:35 APC2:36 CACNA1B:47 OTUD7A:104 MAPK8IP2:111
cerebral.cortex	0.11735641	78	3.622e−04	1.216e−02	KIF1A:31 ADGRB1:53 OPRL1:112 KCNF1:389 DPYSL4:473 RS1:537
EBV.lymphocyte	−0.04125776	576	1.124e−03	1.517e−02	BRCA1:16 ATAD5:36 NIN:38 ACADM:57 NRROS:65 IRAG2:70
testis	−0.03117541	1200	8.459e−04	1.517e−02	NEK5:3 PKD1L3:6 HEATR4:11 RGS22:14 OTOG:56 C11orf16:60
cerebellar.hemisphere	0.04920536	363	1.658e−03	1.791e−02	ADCY1:25 CADM3:32 RNF112:35 SCN2A:81 OTUD7A:104 ARVCF:147
Brodmann.area.9	0.07579939	122	4.108e−03	3.169e−02	FRMPD4:42 TMEM132D:44 GABRB3:78 KCNQ3:172 NETO1:185 CACNG3:266
pancreas	−0.09363665	81	3.745e−03	3.169e−02	FBXW12:73 CPB1:128 TMED6:320 CCDC198:344 PPY:583 RBPJL:601
esophagus.mucosa	0.06084773	175	6.026e−03	4.068e−02	IL36RN:131 FZD10:157 RHOV:203 DSG3:223 TMPRSS11F:247 PGLYRP4:270
vagina	0.08940795	76	7.311e−03	4.387e−02	IL36RN:131 DSG3:223 TMPRSS11F:247 SALL3:264 PGLYRP4:270 ESR1:317
leg.skin	0.05497708	194	9.100e−03	4.467e−02	PARD6G:11 GJB4:34 VSIG8:87 TRPM1:107 IL36RN:131 DSG3:223
skeletal.muscle	0.05580742	190	8.763e−03	4.467e−02	CAVIN4:17 NEB:55 KLHL30:57 ABLIM2:72 PDK4:153 MYBPC2:161
liver	−0.04431114	294	1.031e−02	4.642e−02	DBH:29 ITIH4:106 THPO:166 ACOX2:200 TMPRSS6:230 PIPOX:237
suprapubic.skin	0.05602114	174	1.169e−02	4.854e−02	PARD6G:11 GJB4:34 IL36RN:131 F2RL1:217 DSG3:223 TMPRSS11F:247
substantia.nigra	0.21907682	10	1.650e−02	6.365e−02	TH:254 SLC6A11:742 SLC6A3:1411 RET:1928 FOXB1:2143 SPX:2418
fallopian.tube	−0.09863800	44	2.396e−02	8.624e−02	ADGRG4:110 ADAMTS3:210 CCDC198:344 MMP19:540 CDHR4:618 GAS2L2:629
C1.spinal.cord	0.04732190	134	6.070e−02	2.049e−01	SEC14L5:15 PDE6B:99 PCSK6:215 TTYH2:253 HSPA2:312 ABCA2:381
blood	−0.03392686	235	7.754e−02	2.463e−01	DHX34:104 C3AR1:255 JAK3:291 MPO:373 ARHGAP9:518 NBEAL2:519
nucleus.accumbens	0.05978033	64	9.945e−02	2.827e−01	JAKMIP1:67 INSYN2A:155 OPRK1:625 PNPLA5:862 PENK:968 SHISA7:1002
putamen	0.15126895	10	9.782e−02	2.827e−01	RGS9:26 DIPK1C:166 SLC18A3:1769 OPALIN:2377 ANO3:6651 FAM237A:6651
spleen	−0.02933557	261	1.085e−01	2.931e−01	FCRL3:79 NLRC5:209 FCRL5:277 SLC38A11:311 STAB2:325 LRTM1:340
tibial.artery	0.04939614	85	1.173e−01	3.017e−01	SDK1:143 LMOD1:245 PRELP:278 ADARB1:345 GPR20:510 TLN1:514
hypothalamus	0.06108446	45	1.574e−01	3.863e−01	MC4R:79 PIRT:447 DLX1:521 GABRQ:697 DNER:898 OTP:1362
breast	0.11565003	12	1.657e−01	3.890e−01	TFAP2B:648 KRT5:883 CIDEC:1414 PLIN1:2271 ABCB5:6651 ADIPOQ:6651
minor.salivary.gland	−0.04946394	62	1.796e−01	4.041e−01	MMP27:67 ASCL3:156 MMP7:477 BPIFB1:752 DNASE2B:966 AMTN:1154
greater.omentum	0.07600592	24	1.981e−01	4.278e−01	HAS1:944 FFAR3:1130 CIDEC:1414 BNC1:1499 FAM89A:1882 ADAMTS4:2236
cortex.kidney	−0.03325290	100	2.535e−01	5.264e−01	PKHD1:159 KIF12:177 NPHS2:212 TMEM52B:307 CCDC198:344 NPHS1:687
adrenal.gland	−0.03296587	93	2.746e−01	5.493e−01	SLC16A9:76 DNAH11:203 FDXR:663 SLC14A2:684 ABCC3:773 CELA1:969
Brodmann.area.24	0.06167317	21	3.285e−01	6.335e−01	STUM:60 FAM189A1:346 SHANK1:654 NEUROD6:1822 CSPG5:1962 KCNS1:1988
lung	−0.02423263	120	3.628e−01	6.755e−01	A2M:25 RTKN2:59 ADGRL2:74 SHROOM4:103 ROS1:249 MMP8:656
thyroid	−0.02272489	126	3.821e−01	6.878e−01	SLC12A8:37 ZNF804B:43 SORD:97 TLE6:143 TRMT9B:145 PLA2R1:235
subcutaneous.adipose	−0.03112923	61	4.022e−01	6.972e−01	LVRN:45 PFKFB3:111 KLB:171 GHR:286 GPBAR1:400 ITIH5:456
eye.development	0.03896350	37	4.132e−01	6.972e−01	NRL:410 NOTCH1:482 DLX1:521 SOX8:802 ELF3:1329 SMAD5:1572
caudate.nucleus	0.04649033	19	4.834e−01	7.055e−01	MLC1:294 GPR37L1:1138 PDE1B:1469 ANO3:6651 CHAT:6651 ETNPPL:6651
coronary.artery	−0.06064181	13	4.493e−01	7.055e−01	SUSD5:123 MYO18B:1275 SPINK13:6080 CCDC190:6080 CCL19:6080 COL4A1:6080
endocervix	0.02670030	59	4.797e−01	7.055e−01	TMEM132C:65 SPON2:84 ZNF516:133 ESR1:317 MLPH:594 SNED1:668
esophagogastric.junction	−0.09053778	5	4.833e−01	7.055e−01	F2RL2:1814 ADCY5:6080 CHRM3:6080 COL4A5:6080 GADL1:6080 NA
esophagus.muscularis.mucosa	−0.05159001	18	4.491e−01	7.055e−01	PRUNE2:58 CCDC69:544 RGMB:1241 F2RL2:1814 CHRM3:6080 GADL1:6080
ectocervix	0.03712625	22	5.471e−01	7.707e−01	ESR1:317 ADRA1D:1017 ADAMTSL5:1215 COL5A1:1241 CPXM1:1266 GDF7:6651
transverse.colon	−0.01516668	128	5.566e−01	7.707e−01	TRPM6:12 UCN3:28 MUC4:162 GUCA2B:409 ADAMDEC1:758 SMPDL3B:837
aorta	−0.01352284	128	6.002e−01	8.102e−01	PCDH11X:41 SUSD5:123 FRZB:281 NIBAN1:366 KCNK15:442 VCAN:619